

Geometry (9 th std)

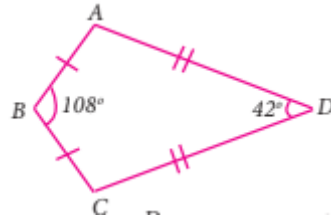
Choose the Appropriate Answer

1. The exterior angle of a triangle is equal to the sum of two

- (a) Exterior angles (b) Interior opposite angles
(c) Alternate angles (d) Interior angles

Ans: (b) Interior opposite angles

2. In the quadrilateral ABCD, $AB = BC$ and $AD = DC$ Measure of $\angle BCD$ is

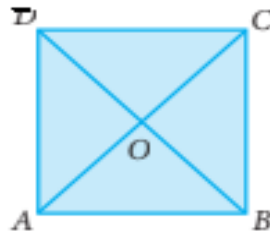


- (a) 150° (b) 30° (c) 105° (d) 72°

Ans: (c) 105°

Join BD: $\angle DBC = 54^\circ$; $\angle BDC = 21^\circ$
 $\therefore \angle BCD = 180^\circ - (54^\circ + 21^\circ) = 180^\circ - 75^\circ = 105^\circ$

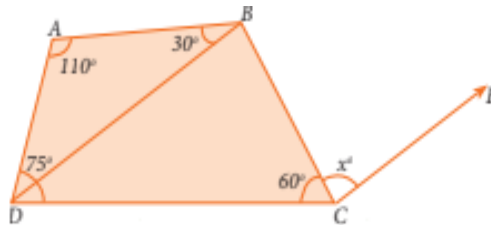
3. ABCD is a square, diagonals AC and BD meet at O. The number of pairs of congruent triangles with vertex O are



- (a) 6 (b) 8 (c) 4 (d) 12

Ans: (a) 6

4. In the given figure $CE \parallel DB$ then the value of x° is

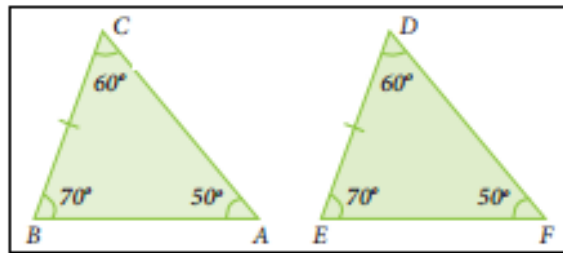


- (a) 45° (b) 30° (c) 75° (d) 85°

Ans: (d) 85°

$\angle B = 360^\circ - (\angle A + \angle D + \angle BCD) = 360^\circ - (110^\circ + 75^\circ + 60^\circ) = 115^\circ$
 $\angle DBC = 115^\circ - 30^\circ = 85^\circ$ (Alternate angles are equal)

5. The correct statement out of the following is



- (a) $\triangle ABC \cong \triangle DEF$ (b) $\triangle ABC \cong \triangle EDF$
 (c) $\triangle ABC \cong \triangle FDE$ (d) $\triangle ABC \cong \triangle FED$

Ans: (d) $\triangle ABC \cong \triangle FED$

6. If the diagonal of a rhombus are equal, then the rhombus is a

- (a) Parallelogram but not a rectangle (b) Rectangle but not a square
 (c) Square (d) Parallelogram but not a square

Ans: (c) Square

7. If bisectors of $\angle A$ and $\angle B$ of a quadrilateral ABCD meet at O, then $\angle AOB$ is

- (a) $\angle C + \angle D$ (b) $12 (\angle C + \angle D)$ (c) $12 \angle C + 13 \angle D$ (d) $13 \angle C + 12 \angle D$

Ans: (b) $12 (\angle C + \angle D)$

8. The interior angle made by the side in a parallelogram is 90° then the parallelogram is a

- (a) Rhombus (b) Rectangle (c) Trapezium (d) Kite

Ans: (b) Rectangle

9. Which one of the following statement is correct?

- (a) Opposite angles of a parallelogram are not equal.
 (b) Adjacent angles of a parallelogram are complementary.
 (c) Diagonals of a parallelogram are always equal.
 (d) Both pairs of opposite sides of a parallelogram are always equal.

Ans: (d) Both pairs of opposite sides of a parallelogram are always equal.

10. The angles of the triangle are $3x - 40$, $x + 20$ and $2x - 10$ then the value of x is ...

- (a) 40° (b) 35° (c) 50° (d) 45°

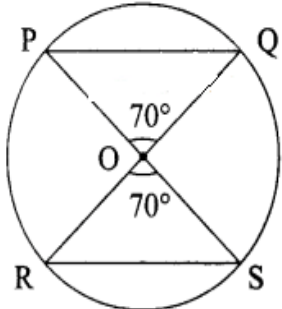
Ans: (b) 35°

$3x - 40 + x + 20 + 2x - 10 = 180^\circ \text{ (Sum of the angles of a triangle is } 180^\circ)$ $6x - 30 = 180^\circ \Rightarrow 6x = 180^\circ + 30^\circ \Rightarrow x = 210^\circ / 6 = 35^\circ$

11. PQ and RS are two equal chords of a circle with centre O such that $\angle POQ = 70^\circ$, then $\angle ORS =$

- (a) 60° (b) 70° (c) 55° (d) 80°

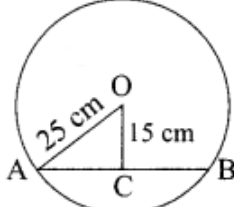
Ans: (c) 55°



$\angle POQ = 70^\circ$ (Vertically opposite Angle)
 $\angle ORS$ and $\angle OSR$ are equal. (OR = OS radius of the circle)
 $\angle ORS + \angle OSR + \angle ROS = 180^\circ$
 $x^\circ + x^\circ + 70^\circ = 180^\circ \Rightarrow 2x + 70^\circ = 180^\circ \Rightarrow 2x = 180^\circ - 70^\circ = 110^\circ$
 $x = 110^\circ / 2 = 55^\circ \Rightarrow \therefore \angle ORS = 55^\circ$

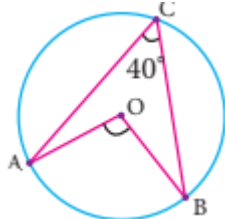
12. A chord is at a distance of 15 cm from the centre of the circle of radius 25 cm. The length of the chord is
- (a) 25 cm (b) 20 cm (c) 40 cm (d) 18 cm

Ans: (c) 40 cm



In the right triangle OAC, $AC^2 = OA^2 - OC^2 = 25^2 - 15^2$
 $AC^2 = 400$ $AC = 20$
 Length of the chord AB = $20 + 20 = 40$ cm.

13. In the figure, O is the centre of the circle and $\angle ACB = 40^\circ$ then $\angle AOB = \dots\dots\dots$



- (a) 80° (b) 85° (c) 70° (d) 65°

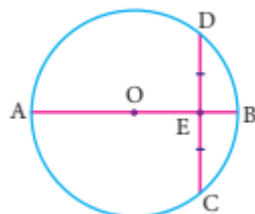
Ans: (a) 80°

14. In a cyclic quadrilateral ABCD, $\angle A = 4x$, $\angle C = 2x$ the value of x is
- (a) 30° (b) 20° (c) 15° (d) 25

Ans: (a) 30°

$\angle A + \angle C = 180^\circ$ (Sum of the opposite angle of cyclic quadrilateral is 180°)
 $4x + 2x = 180^\circ \Rightarrow x = 180^\circ / 6 = 30^\circ$

15. In the figure, O is the centre of a circle and diameter AB bisects the chord CD at a point E such that $CE = ED = 8$ cm and $EB = 4$ cm. The radius of the circle is ...



- (a) 8 cm (b) 4 cm (c) 6 cm (d) 10 cm

Ans: (a) 10 cm

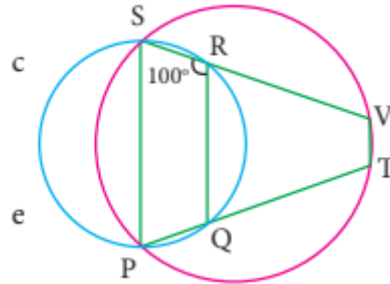
Let the radius OD be x.

$OE = OB - BE = x - 4$ (OB radius of the circle)

In the $\triangle OED$,

$$OD^2 = OE^2 + ED^2 \Rightarrow x^2 = (x - 4)^2 + 8^2 \quad x^2 = x^2 + 16 - 8x + 64 \Rightarrow 8x = 80 \Rightarrow x = 10 \text{ cm}$$

16. In the figure, PQRS and PTVS are two cyclic quadrilaterals, If $\angle QRS = 80^\circ$, then $\angle TVS = \dots\dots\dots$



- (a) 80° (b) 100° (c) 70° (d) 90°

Ans: (a) 80°

$$\angle SPQ = 180^\circ - 80^\circ = 100^\circ$$

$$\text{In the cyclic quadrilateral PTVS, } \angle V = 180^\circ - 100^\circ = 80^\circ$$

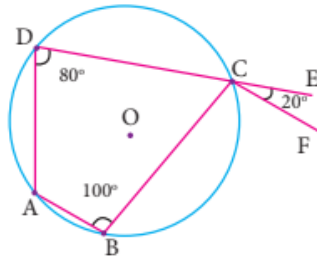
17. If one angle of a cyclic quadrilateral is 75° , then the opposite angle is $\dots\dots\dots$

- (a) 100° (b) 105° (c) 85° (d) 90°

Ans: (b) 105°

$$\text{Opposite angles} = 180^\circ - 75^\circ = 105^\circ \text{ (Sum of the opposite angle is } 180^\circ)$$

18. In the figure, ABCD is a cyclic quadrilateral in which DC produced to E and CF is drawn parallel to AB such that $\angle ADC = 80^\circ$ and $\angle ECF = 20^\circ$, then $\angle BAD = ?$



- (a) 100° (b) 20° (c) 120° (d) 110°

Ans: (c) 120°

$$\angle BAD = \angle ABC + \angle ECF = 100^\circ + 20^\circ = 120^\circ$$

19. AD is a diameter of a circle and AB is a chord. If $AD = 30 \text{ cm}$ and $AB = 24 \text{ cm}$ then the distance of AB from the centre of the circle is $\dots\dots\dots$

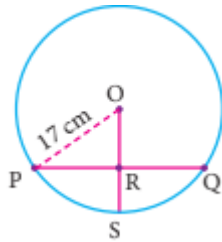
- (a) 10 cm (b) 9 cm (c) 8 cm (d) 6 cm

Ans: (b) 9 cm

$$\text{In } \triangle AOC, AO = 15 \text{ cm, } AC = 12 \Rightarrow AB = 2 \times AC = 2 \times 12 = 24 \text{ cm}$$

$$\text{In } \triangle AOC, OC^2 = AO^2 - AC^2 = 15^2 - 12^2 = 225 - 144 = 81 \Rightarrow OC = \sqrt{81} = 9 \text{ cm}$$

20. In the given figure, If $OP = 17$ cm, $PQ = 30$ cm and OS is perpendicular to PQ , then RS is



- (a) 10 cm (b) 6 cm (c) 7 cm (d) 9 cm

Ans: (d) 9 cm

In $\triangle OPR$, $OP = 17$ cm, $PR = 30 / 2 = 15$ cm
 $OR^2 = OP^2 - PR^2 = 17^2 - 15^2 = (17 + 15)(17 - 15) = 32 \times 2 \Rightarrow OR^2 = \sqrt{64} = 8$ cm
 $RS = OS - OR = 17 - 8 = 9$ cm

21. The angle sum of a convex polygon with number of sides 7 is

- (a) 900° (b) 1080° (c) 1444° (d) 720°

Ans: (a) 900°

22. What is the name of a regular polygon of six sides?

- (a) Square (b) Equilateral triangle
 (c) Regular hexagon (d) Regular octagon

Ans: (c) Regular hexagon

23. One angle of a parallelogram is a right angle. The name of the quadrilateral is

- (a) Square (b) Rectangle (c) Rhombus (d) Kite

Ans: (b) Rectangle

24. If all the four sides of a parallelogram are equal and the adjacent angles are of 120° and 60° , then the name of the quadrilateral is

- (a) Rectangle (b) Square (c) Rhombus (d) Kite

Ans: (c) Rhombus

25. In a parallelogram $\angle A : \angle B = 1 : 2$. Then $\angle A$

- (a) 30° (b) 60° (c) 45° (d) 90°

Ans: (b) 60°

26. Which of the following is a formula to find the sum of interior angles of a quadrilateral of w -sides?

- (a) $n / 2 \times 180^\circ$ (b) $(n+1) / 2 \times 180^\circ$
 (c) $(n-1) / 2 \times 180^\circ$ (d) $(n-2) \times 180^\circ$

Ans: (d) $(n-2) \times 180^\circ$

27. Diagonal of which of the following quadrilaterals do not bisect it into two congruent triangles?

- (a) Rhombus (b) Trapezium (c) Square (d) Rectangle

Ans: (b) Trapezium

28. The point of concurrency of the medians of a triangle is known as

- (a) Circumcentre (b) Incentre (c) Orthocentre (d) Centroid

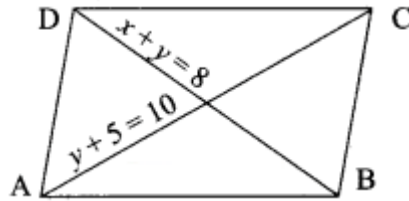
Ans: (d) Centroid

29. Orthocentre of a triangle is the point of concurrency of

- (a) Medians (b) Altitudes
(c) Angle bisectors (d) Perpendicular bisectors of side

Ans: (b) Altitudes

30. ABCD is a parallelogram as shown in below figure. Find x and y.



- (a) 1, 7 (b) 2, 6 (c) 3, 5 (d) 4, 4

Ans: (c) 3, 5

31. A circle divides the plane into part

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (c) 3

32. The longest chord of a circle is a

- (a) Radius (b) Diameter (c) Chord (d) Secant

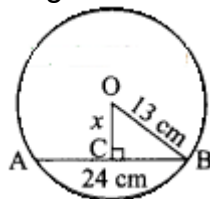
Ans: (b) Diameter

33. Opposite angles of a cyclic quadrilateral are

- (a) Supplementary (b) Complementary (c) Equal (d) None of these

Ans: (a) Supplementary

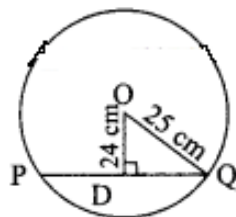
34. The value of x from figure is if 'O' is the centre of the circle



- (a) 20 cm (b) 15 cm (c) 12 cm (d) 5 cm

Ans: (d) 5 cm

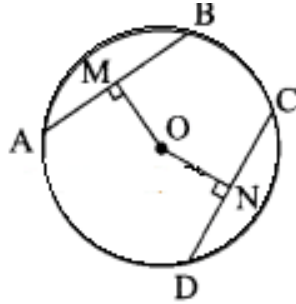
35. If PQ = x and 'O' is the centre of the circle, then x=



- (a) 7 cm (b) 14 cm (c) 8 cm (d) 13 cm

Ans: (b) 14 cm

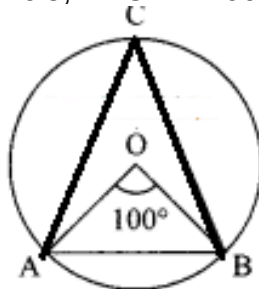
36. In figure $OM = ON = 8\text{ cm}$ and $AB = 30\text{ cm}$, then $CD = \dots\dots\dots$



- (a) 15 cm (b) 30 cm (c) 40 cm (d) 10 cm

Ans: (b) 30 cm

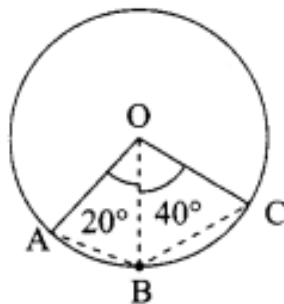
37. O is the centre of a circle, $\angle AOB = 100^\circ$. Then angle $\angle ACB = \dots\dots\dots$



- (a) 80° (b) 40° (c) 50° (d) 60°

Ans: (c) 50°

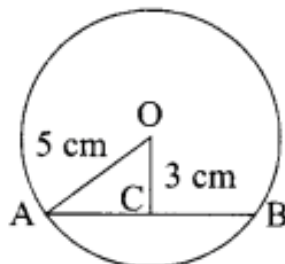
38. In a circle, with centre O, $\angle AOB = 20^\circ$, $\angle BOC = 40^\circ$, arc $BC = 4$. Then length of arc AB will be $\dots\dots\dots$



- (a) 8 cm (b) 6 cm (c) 2 cm (d) 1 cm

Ans: (c) 2 cm

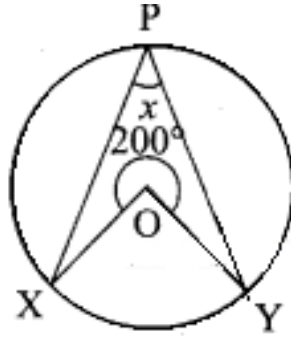
39. In the figure, $OC = 3\text{ cm}$ and radius of circle is 5 cm . Then $AB = \dots\dots\dots$



- (a) 4 cm (b) 5 cm (c) 6 cm (d) 8 cm

Ans: (d) 8 cm

40.O is the centre of the circle. The value of x in the given diagram is



- (a) 100° (b) 160° (c) 200° (d) 80°

Ans: (d) 80°

41.A line joining two endpoints is called:

- (a) Line segment (b) A ray (c) Parallel lines (d) Intersecting lines

Ans: (a) Line segment

42.An acute angle is:

- (a) More than 90 degrees (b) Less than 90 degrees
(c) Equal to 90 degrees (d) Equal to 180 degrees

Ans: (b) Less than 90 degrees

43.A reflex angle is:

- (a) More than 90 degrees (b) Equal to 90 degrees
(c) More than 180 degrees (d) Equal to 180 degrees

Ans: (c) More than 180 degrees

44.A straight angle is equal to:

- (a) 0° (b) 90° (c) 180° (d) 360°

Ans: (c) 180°

45.Two angles whose sum is equal to 180° are called:

- (a) Vertically opposite angles (b) Complementary angles
(c) Adjacent angles (d) Supplementary angles

Ans: (d) Supplementary angles

46.Intersecting lines cut each other at:

- (a) One point (b) Two points (c) Three point's (d) Null

Ans: (a) One point

47.Two parallel lines intersect at:

- (a) One point (b) Two points (c) Three points (d) Null

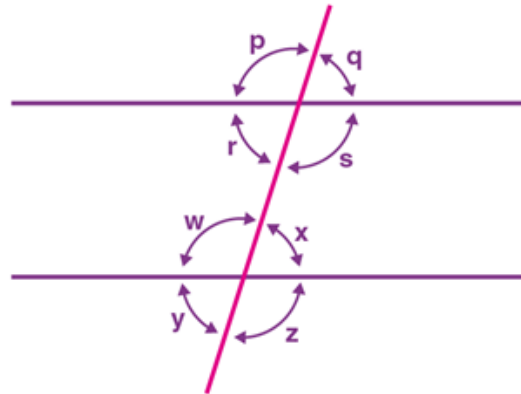
Ans: (d) Null

48.If two lines intersect each other, then the vertically opposite angles are:

- (a) Equal (b) Unequal (c) Can't be determined (d) None of the above

Ans: (a) Equal

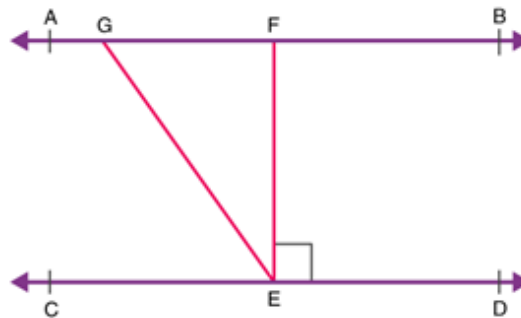
49. In the figure below, which of the following are corresponding angle pairs?



- (a) $\angle p$ and $\angle q$ (b) $\angle p$ and $\angle w$ (c) $\angle p$ and $\angle x$ (d) $\angle p$ and $\angle z$

Ans: (b) $\angle p$ and $\angle w$

50. If $AB \parallel CD$, $EF \perp CD$ and $\angle GED = 135^\circ$ as per the figure given below. The value of $\angle AGE$ is:



- (a) 120° (b) 140° (c) 90° (d) 135°

Ans: (d) 135°

51. An exterior angle of a triangle is 105° and its two interior opposite angles are equal. Each of these equal angles is

- (a) $37\frac{1}{2}^\circ$ (b) $72\frac{1}{2}^\circ$ (c) 75° (d) $52\frac{1}{2}^\circ$

Ans: (d) $52\frac{1}{2}^\circ$

$$105^\circ = x + x \Rightarrow 2x = 105^\circ \Rightarrow x = 52\frac{1}{2}^\circ$$

52. If two interior angles on the same side of a transversal intersecting two parallel lines are in the ratio 2 : 3, then the greater of the two angles is:

- (a) 54° (b) 108° (c) 120° (d) 136°

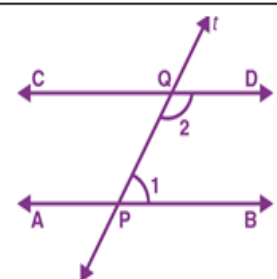
Ans: (b) 108°

$$\angle 1 : \angle 2 = 2 : 3 \Rightarrow \text{Let } \angle 1 = 2x \text{ and } \angle 2 = 3x.$$

$$\angle 1 + \angle 2 = 180^\circ \Rightarrow \angle 1 = 2x \text{ and } \angle 2 = 3x \Rightarrow 2x + 3x = 180^\circ$$

$$5x = 180^\circ \Rightarrow x = 180^\circ / 5 = 36^\circ \Rightarrow 3x > 2x. \text{ It means } \angle 2 > \angle 1$$

$$\text{The value of } \angle 2 = 3(36^\circ) = 108^\circ$$



53.If one angle of a triangle is equal to the sum of the other two angles, then the triangle is

- (a) a right triangle (b) an isosceles triangle
(c) an equilateral triangle (d) an obtuse triangle

Ans: (a) a right triangle

54.The angles of a triangle are in the ratio 5: 3: 7. The triangle is

- (a) a right triangle (b) an acute-angled triangle
(c) an obtuse-angled triangle (d) an isosceles triangle

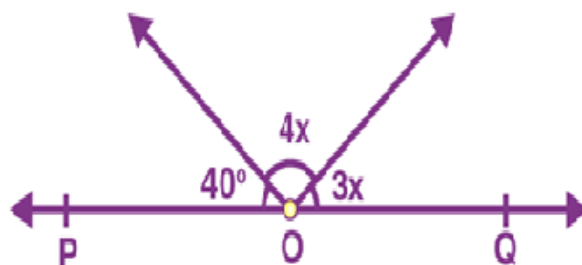
Ans: (b) an acute-angled triangle

55.Angles of a triangle are in the ratio 2: 4: 3. The smallest angle of the triangle is

- (a) 20° (b) 40° (c) 60° (d) 80°

Ans: (b) 40°

56.In the given figure, POQ is a line. The value of x is



- (a) 20° (b) 25° (c) 30° (d) 35°

Ans: (a) 20°

$$POQ = 180^\circ \Rightarrow 40^\circ + 4x + 3x = 180^\circ \Rightarrow 7x = 180^\circ - 40^\circ \Rightarrow x = 20^\circ$$

57.An obtuse angle is

- (a) Less than 90° (b) Greater than 90°
(c) Equal to 90° (d) Equal to 180°

Ans: (b) Greater than 90°

58.In triangle ABC, if $AB=BC$ and $\angle B = 70^\circ$, $\angle A$ will be:

- (a) 70° (b) 110° (c) 55° (d) 130°

Ans: (c) 55°

$$AB = BC \Rightarrow \angle A = \angle C \text{ and } \angle B = 70^\circ \Rightarrow \angle A + \angle B + \angle C = 180^\circ \\ 2\angle A + \angle B = 180^\circ \Rightarrow 2\angle A = 180 - \angle B = 180 - 70 = 110^\circ \Rightarrow \angle A = 55^\circ$$

59.For two triangles, if two angles and the included side of one triangle are equal to two angles and the included side of another triangle. Then the congruency rule is:

- (a) SSS (b) ASA (c) SAS (d) None of the above

Ans: (b) ASA

60.A triangle in which two sides are equal is called _____ triangle.

- (a) Scalene (b) Equilateral (c) Isosceles (d) None of the above

Ans: (c) Isosceles

61. The angles opposite to equal sides of a triangle are _____ angles.
(a) Equal (b) Unequal (c) supplementary (d) Complementary

Ans: (a) Equal

62. If E and F are the midpoints of equal sides AB and AC of a triangle ABC. Then:
(a) $BF = AC$ (b) $BF = AF$ (c) $CE = AB$ (d) $BF = CE$

Ans: (d) $BF = CE$

63. ABC is an isosceles triangle in which altitudes BE and CF are drawn to equal sides AC and AB, respectively. Then:

(a) $BE > CF$ (b) $BE < CF$ (c) $BE = CF$ (d) None of the above

Ans: (c) $BE = CF$

64. If ABC and DBC are two isosceles triangles on the same base BC. Then:

(a) $\angle ABD = \angle ACD$ (b) $\angle ABD > \angle ACD$
(c) $\angle ABD < \angle ACD$ (d) None of the above

Ans: (a) $\angle ABD = \angle ACD$

65. If ABC is an equilateral triangle, then each angle equals to:

(a) 90° (b) 180° (c) 120° (d) 60°

Ans: (d) 60°

66. If AD is an altitude of an isosceles triangle ABC in which $AB = AC$. Then:

(a) $BD = CD$ (b) $BD > CD$ (c) $BD < CD$ (d) None of the above

Ans: (a) $BD = CD$

67. In a right triangle, the longest side is:

(a) Perpendicular (b) Hypotenuse (c) Base (d) None of the above

Ans: (b) Hypotenuse

68. Which of the following is not a criterion for congruence of triangles?

(a) SAS (b) ASA (c) SSA (d) SSS

Ans: (c) SSA

69. In triangles ABC and PQR, $AB = AC$, $\angle C = \angle P$ and $\angle B = \angle Q$. The two triangles are

(a) Isosceles and congruent (b) Isosceles but not congruent
(c) Congruent but not isosceles (d) Neither congruent nor isosceles

Ans: (b) Isosceles but not congruent

70. In $\triangle PQR$, $\angle R = \angle P$ and $QR = 4$ cm and $PR = 5$ cm. Then the length of PQ is

(a) 2 cm (b) 2.5 cm (c) 4 cm (d) 5 cm

Ans: (c) 4 cm

71. If $AB = QR$, $BC = PR$ and $CA = PQ$, then

(a) $\triangle PQR \cong \triangle BCA$ (b) $\triangle BAC \cong \triangle RPQ$
(c) $\triangle CBA \cong \triangle PRQ$ (d) $\triangle ABC \cong \triangle PQR$

Ans: (c) $\triangle CBA \cong \triangle PRQ$

72. If $\triangle ABC \cong \triangle PQR$, then which of the following is not true?

- (a) $AC = PR$ (b) $BC = PQ$ (c) $QR = BC$ (d) $AB = PQ$
Ans: (b) $BC = PQ$

73. In $\triangle ABC$, $BC = AB$ and $\angle B = 80^\circ$. Then $\angle A$ is equal to
(a) 40° (b) 50° (c) 80° (d) 100°

Ans: (b) 50°

In a triangle, ABC, $BC = AB$ and $\angle B = 80^\circ$. $x + 80^\circ + x = 180^\circ \Rightarrow 2x + 80^\circ = 180^\circ \Rightarrow 2x = 100^\circ \Rightarrow x = 50^\circ$
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74. Two sides of a triangle are of lengths 5 cm and 1.5 cm. The length of the third side of the triangle cannot be
(a) 3.4 cm (b) 3.6 cm (c) 3.8 cm (d) 4.1 cm

Ans: (a) 3.4 cm

75. In $\triangle ABC$, $AB = AC$ and $\angle B = 50^\circ$. Then $\angle C$ is equal to
(a) 40° (b) 50° (c) 80° (d) 130°

Ans: (b) 50°

76. The sum of all the angles of a quadrilateral is equal to:
(a) 180° (b) 270° (c) 360° (d) 90°

Ans: (c) 360°

77. A trapezium has:

- (a) One pair of opposite sides parallel
(b) Two pairs of opposite sides parallel to each other
(c) All its sides are equal
(d) All angles are equal

Ans: (a) One pair of opposite sides parallel

78. A rhombus can be a:

- (a) Parallelogram (b) Trapezium (c) Kite (d) Square

Ans: (d) Square

79. A diagonal of a parallelogram divides it into two congruent:

- (a) Square (b) Parallelogram (c) Triangles (d) Rectangle

Ans: (c) Triangles

80. In a parallelogram, opposite angles are:

- (a) Equal (b) Unequal (c) Can't determine (d) None of these

Ans: (a) Equal

81. The diagonals of a parallelogram:

- (a) Equal (b) Unequal (c) Bisect each other (d) Have no relation

Ans: (c) Bisect each other

82. Each angle of the rectangle is:

- (a) More than 90° (b) Less than 90° (c) Equal to 90° (d) Equal to 45°

Ans: (c) Equal to 90°

83. The angles of a quadrilateral are in the ratio 4: 5: 10: 11. The angles are:

- (a) $36^\circ, 60^\circ, 108^\circ, 156^\circ$ (b) $48^\circ, 60^\circ, 120^\circ, 132^\circ$
(c) $52^\circ, 60^\circ, 122^\circ, 126^\circ$ (d) $60^\circ, 60^\circ, 120^\circ, 120^\circ$

Ans: (b) $48^\circ, 60^\circ, 120^\circ, 132^\circ$

$$4x + 5x + 10x + 11x = 360^\circ \Rightarrow 30x = 360^\circ \Rightarrow x = 12^\circ$$
$$\text{Hence, angles are } 4x = 48^\circ, 5x = 60^\circ, 10x = 120^\circ, 11x = 132^\circ$$

84. If ABCD is a trapezium in which $AB \parallel CD$ and $AD = BC$, then:

- (a) $\angle A = \angle B$ (b) $\angle A > \angle B$ (c) $\angle A < \angle B$ (d) None of the above

Ans: (a) $\angle A = \angle B$

85. Which of the following is not true for a parallelogram?

- (a) Opposite sides are equal
(b) Opposite angles are equal
(c) Opposite angles are bisected by the diagonals
(d) Diagonals bisect each other.

Ans: (c) Opposite angles are bisected by the diagonals

86. Three angles of a quadrilateral are $75^\circ, 90^\circ$ and 75° . The fourth angle is

- (a) 90° (b) 95° (c) 105° (d) 120°

Ans: (d) 120°

$$75^\circ + 90^\circ + 75^\circ + x = 360^\circ \Rightarrow x = 360^\circ - 240^\circ = 120^\circ$$

87. The quadrilateral formed by joining the mid-points of the sides of a quadrilateral PQRS, taken in order, is a rhombus, if

- (a) PQRS is a rhombus
(b) PQRS is a parallelogram
(c) Diagonals of PQRS are perpendicular
(d) Diagonals of PQRS are equal.

Ans: (d) Diagonals of PQRS are equal.

88. The center of the circle lies in _____ of the circle.

- (a) Interior (b) Exterior (c) Circumference (d) None of the above

Ans: (a) Interior

89. If there are two separate circles drawn apart from each other, then the maximum number of common points they have:

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (a) 0

90. Which of these angles we cannot construct it using a ruler and compasses?

- (a) 120° (b) 70° (c) 60° (d) All can be constructed

Ans: (b) 70°

91. Which of these angles cannot be constructed using a ruler and compasses?

- (a) 120° (b) 60° (c) 140° (d) 135°

Ans: (c) 140°

92. If a , b and c are the lengths of three sides of a triangle, then:
(a) $a + b > c$ (b) $a - b > c$ (c) $a + b = c$ (d) $a - b = c$
Ans: (a) $a + b > c$

93. To construct a bisector of a given angle, we need:
(a) A ruler (b) A compass (c) A protractor (d) Both ruler & compass
Ans: (d) Both ruler & compass

94. To construct an angle of 60 degrees, we need to draw first:
(a) A ray (b) An arc (c) Two rays (d) A straight line
Ans: (a) A ray

95. If we want to construct a triangle, given its perimeter, then we need to know:
(a) Sum of two sides of triangle
(b) Difference between two sides of triangle
(c) One base angles
(d) Two base angles
Ans: (d) Two base angles

96. The construction of a triangle ABC in which $AB = 4$ cm, $\angle A = 60^\circ$ is not possible when the difference of BC and AC is equal to
(a) 2.5 cm (b) 3 cm (c) 3.5 cm (d) 4.5 cm
Ans: (d) 4.5 cm

Among the options given, $4.5 > 4$, since the difference of any two sides of a triangle cannot be greater than the third side.

97. A triangle ABC with $AB = 4$ cm and $\angle A = 60^\circ$ and $\angle B = 40^\circ$ is constructed. Then what is the measurement of $\angle C$?
(a) 40° (b) 60° (c) 80° (d) 100°
Ans: (c) 80°

98. The construction of a triangle ABC, given that $BC = 6$ cm, $\angle B = 45^\circ$ is not possible when the difference of AB and AC is equal to:
(a) 4 cm (b) 5 cm (c) 5.2 cm (d) 6.9 cm
Ans: (d) 6.9 cm

Among the options given, $6.9 > 6$, since the difference of any two sides of a triangle cannot be greater than the third side.

99. The internal and external bisectors of an angle forms ____ angle.
(a) Acute (b) Right (c) Obtuse (d) Straight
Ans: (b) Right

100. The bisector of an angle lies in its _____ plane.
(a) Interior (b) Exterior (c) Anywhere of the (d) On the arms of the
Ans: (a) Interior